

AMIRREZA VALAEI

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RESEARCH INTERESTS

- Reinforcement Learning
- Statistics
- Optimization

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Daejeon, South Korea

- M.Sc. in **Electrical Engineering**, U-AIM Lab
- *Advisor*: Prof. Chang D. Yoo

Sep 2026 – Jun 2028 (*expected*)

Sharif University of Technology

Tehran, Iran

- B.Sc. in **Electrical Engineering** — GPA: **17.65/20 (3.7/4.0)**
- Minor in **Applied Mathematics**
- **B.Sc. Thesis**: Numerical Optimization in TRPO

Sep 2021 – Jun 2026

Feb 2023 – Present

Prof. [Hamed Shah-Mansouri](#)

Allameh Jafari High School (NODET)

Marand, Iran

- **Mathematics and Physics** — GPA: **19.23/20**

2018 – 2021

PUBLICATIONS & PREPRINTS

- **Controlling Gender Bias in Retrieval** In Proceedings of ECIR 2026 [arXiv](#)
University of Amsterdam
Authors: Amirabbas Afzali*, **Amirreza Valaei***, Iman Ahmadi, Mohammad Aliannejadi (*Equal contribution)
Converted the Backpack LM into an encoder, enabling bias mitigation for ranking tasks.
- **Second-Order Policy Gradient Method for the LQR** Engineering Applications of AI [arXiv](#)
Max Planck Institute for Software Systems
Authors: **Amirreza Valaei**, Arash Bahari Kordabad, Sadegh Soudjani
Designed an algorithm for exact second-order updates in deterministic policy-gradient methods for LQR.

OTHER RESEARCH EXPERIENCE

Sharif University of Technology

- **Numerical Optimization in Trust Region Policy Optimization** Oct 2024 – Aug 2025
Supervisor: Prof. [Hamed Shah-Mansouri](#)
Analyzed conjugate-gradient updates in TRPO; reproduced baselines and ran controlled ablations.
- **Robust Reinforcement Learning** Apr 2025 – Jan 2026
Supervisor: Prof. [Sajjad Amini](#)
Led a small research group on robust RL, conducted literature reviews, and implemented algorithms.

SELECTED ADVANCED COURSES

- **Undergraduate Courses**: Linear Algebra – **19.8/20**, Machine Learning – **19.5/20**, Signals and Systems – **19.2/20**, Convex Optimization – **17/20**, Game Theory – **17.8/20**, Numerical Optimization – **19/20**, Mathematical Analysis I – **18/20**, Differential Geometry – **18/20**, Mathematical Analysis II – **18.5/20**, Stochastic Processes – **19.2/20**
 - **Graduate Courses**: Deep Learning – **17.2/20**, Deep Reinforcement Learning – **20/20**, Convex Optimization II – **18.7/20**, High-Dimensional Probability Analysis – **19.3/20**, Online Learning[†], Optimization Seminar – **18/20**, Bandit Algorithms[†], Information Theory – Statistics – and Learning – **19.5/20**, Optimal Control – **15.2/20**, Probabilistic Graphical Models – **19.3/20**, *Deep Generative Models*[†], *Neuroscience*[†]
- [†] Audited or self-studied *Italicized courses indicate current enrollment or study.*

SELECTED ADVANCED COURSE PROJECTS

- Poster presentation on "Robust Reinforcement Learning" and related projects [[code and report](#)]
- Implementation of the GANBERT model for a text classification task [[code and report](#)]
- Presentation on online geometric optimization in the bandit setting against an adaptive adversary [[slides](#)]
- High-dimensional analysis of the Neural Tangent Kernel [[slides and report](#)]
- Expectation-Maximization for a mixture of Gaussians [[code and report](#)]
- Lower-bound for non-stationary bandit algorithm [[slides](#)]
- Multiple deep learning course projects [[code](#)]

TEACHING EXPERIENCE

Teaching Assistant, Sharif University of Technology (*More information is available on my [teaching page](#)*)

- **Undergraduate Courses:** Probability and Statistics (Sp'23, Sp'24, Fa'24), Signals and Systems (Fa'23, Sp'24), Linear Algebra (Fa'23, Sp'24), Machine Learning (Fa'23, Sp'24, Fa'24 [Lead TA](#), Sp'25, Fa'25), Convex Optimization I (Fa'24, Sp'25), Reinforcement Learning (Fa'25)
- **Graduate Courses:** Deep Learning (Fa'24), Deep Reinforcement Learning (Sp'25) High-Dimensional Probability Analysis (Fa'25) High-Dimensional Statistics (Fa'25)

VOLUNTEER EXPERIENCE

- **Scientific Associate Principal**, Student Association, Sharif University ([Resana](#)) Jun 2023 – Jul 2024
Resana is the scientific community of the Electrical Engineering Department at Sharif University of Technology.
- **Organizer**, First AIVengers Summer School Sep 2023
Planned and managed a global AI/ML student event featuring Ph.D. tutors from leading institutions.
- **Central Council Member**, [ReACT 2024](#) Dec 2024
Helped organize a large-scale event with 1,500+ attendees and internationally recognized researchers.
- **TEDx Sharif University of Technology** [TEDxSUT](#) Nov 2025
Assisted in organizing and managing the TEDx event at Sharif University of Technology.

WORK EXPERIENCE

[Tabdeal](#)

Aug 2024 – Jan 2025

Software Engineering Intern (On-site; Tehran, Iran)

- Assisted in backend Python development, optimized database search queries, and integrated internal information retrieval APIs for customer support tracking tools.

SKILLS AND INTERESTS

- **Programming:** Python (PyTorch, OpenCV, LangChain, Django), MATLAB, C++, Java, MySQL
- **Languages:** Persian, English, Turkish, Russian (Elementary), German (Beginner)
- **Interests:** Chess, Cinema, Soccer, Literature

HONORS & CERTIFICATIONS

- **Ranked 51st/165,000** in the Mathematics & Physics University Entrance Exam 2021
- **Ranked 3rd** in the Control Systems cohort of Electrical Engineering at Sharif University 2026
- IELTS 7.5 (L:8, R:8, W:6.5, S:7) ([verification link](#)) 2025